FORM 2

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COMPLETE SPECIFICATION

(see section 10)

Adaptive Capability Reallocation System for Intelligent Railway Operational Safety

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The following specification particularly describes the invention and the manner in which it is to be performed

FIELD OF INVENTION

The present invention relates generally to intelligent railway operation and safety systems. More particularly, the present invention relates to a computer-implemented system and method for adaptive capability reallocation among distributed intelligent railway agents using digital twin technology, capability health assessment, dynamic capability reassignment, safety validation, and rollback recovery mechanisms to maintain resilient and uninterrupted railway operations during partial capability degradation

BACKGROUND OF THE INVENTION

Modern railway transportation systems increasingly employ intelligent software agents, digital twin technology, artificial intelligence, Internet of Things (IoT), and distributed computing architectures to perform safety-critical functions including railway signalling, train tracking, route planning, electronic interlocking, collision avoidance, and operational monitoring. Although such technologies improve operational efficiency, conventional railway systems generally assign fixed operational responsibilities to individual intelligent agents, making the overall system vulnerable to partial degradation of individual software capabilities.

Chinese Patent CN119885877A, entitled Rail Transit Equipment Monitoring System Based on Multi-Source Fusion Data Management and Digital Twin, discloses a digital twin-based railway monitoring architecture employing multi-source data fusion for infrastructure monitoring, fault diagnosis, and predictive maintenance. However, the disclosed system primarily focuses on monitoring railway infrastructure and does not provide selective redistribution of degraded railway operational capabilities among intelligent agents during runtime or capability-level adaptive reconfiguration.

Chinese Patent CN119809054B, entitled Digital Twin Management System and Method for Railway Construction, discloses a digital twin framework for railway construction management, construction monitoring, and lifecycle optimization. However, the disclosed system is directed toward construction management rather than real-time railway operation and neither performs capability-level health assessment nor dynamically reallocates degraded operational capabilities among intelligent railway agents while validating operational safety before deployment.

Existing railway safety solutions predominantly rely on complete software-agent replacement, static redundancy, or conventional failover mechanisms after fault detection. Such approaches generally interrupt railway operations, increase recovery time, consume additional computational resources, and fail to distinguish between healthy and degraded operational capabilities within an intelligent agent. Consequently, even when only a specific operational capability becomes degraded, the entire intelligent agent is unnecessarily replaced or restarted, thereby reducing operational resilience and computational efficiency. Furthermore, existing systems do not provide integrated mechanisms for capability-level health assessment, adaptive capability reassignment, dependency-aware validation, railway-specific safety verification, and automatic rollback following unsuccessful capability migration.

Accordingly, there exists a need for an improved computer-implemented system and method capable of continuously assessing individual operational capabilities of intelligent railway agents, selectively reallocating only degraded capabilities to compatible agents, validating the reconfigured operational state against predefined railway safety constraints, and automatically restoring previously validated configurations when required, thereby maintaining uninterrupted, resilient, and safety-compliant railway operations under partial system degradation. The present invention addresses these technical problems by providing an adaptive capability reallocation system for intelligent railway operational safety.

OBJECTIVE OF THE INVENTION

The primary objective of the present invention is to provide an adaptive capability reallocation system for intelligent railway operational safety that continuously maintains safe and uninterrupted railway operations by selectively redistributing degraded operational capabilities among multiple intelligent railway agents instead of replacing entire software agents. The invention seeks to overcome the limitations of existing railway safety systems that rely on static redundancy, complete agent replacement, or conventional failover mechanisms, which often increase recovery time, interrupt operations, and inefficiently utilize computational resources.

Another objective of the present invention is to implement a capability-centric health assessment mechanism configured to continuously evaluate the operational condition of individual railway capabilities using multiple runtime parameters including communication quality, sensor reliability, computational performance, execution latency, resource availability, and prediction confidence.

By independently assessing each operational capability rather than the overall software agent, the invention enables early identification of partial degradation before complete system failure.

A further objective of the invention is to provide an adaptive capability reallocation mechanism configured to identify degraded railway operational capabilities, determine compatible intelligent railway agents using capability compatibility, dependency relationships, operational health, and resource availability, and dynamically reassign only the affected capabilities while allowing unaffected operational functions to continue uninterrupted.

Another objective of the invention is to incorporate a railway-specific safety validation mechanism that verifies every proposed capability reassignment against predefined railway operational constraints including route consistency, signal compatibility, interlocking conditions, track occupancy, collision safety, and operational dependencies before activating the migrated capability, thereby preventing unsafe system reconfiguration.

Another objective of the invention is to provide an automated rollback and recovery mechanism configured to restore previously validated capability assignments whenever a migrated capability fails to satisfy predefined operational or safety requirements after deployment. This feature ensures continuous railway operation while preventing unsafe operational states during adaptive system reconfiguration.

A further objective of the invention is to integrate railway data acquisition, digital twin modelling, intelligent multi-agent coordination, capability health assessment, adaptive capability reallocation, railway safety validation, rollback recovery, and operational monitoring into a unified computer-implemented framework capable of improving the resilience, reliability, scalability, and fault tolerance of intelligent railway operation under dynamic operating conditions and partial system degradation.

SUMMARY OF THE INVENTION

The following summary is provided to facilitate a clear understanding of the new features in the disclosed embodiment and it is not intended to be a full, detailed description. A detailed description of all the aspects of the disclosed invention can be understood by reviewing the full specification, the drawing and the claims and the abstract, as a whole.

In one aspect, the present invention provides a computer-implemented system for adaptive capability reallocation in intelligent railway operational safety, wherein individual operational capabilities executed by distributed intelligent railway agents are continuously monitored, selectively reassigned upon degradation, and validated prior to deployment to maintain uninterrupted railway operations.

In one aspect, the system comprises a railway data acquisition module configured to continuously collect operational information from railway infrastructure including signalling systems, point machines, track circuits, train positioning systems, communication networks, voltage monitoring units, and environmental sensors, and to generate synchronized operational data representative of the current railway environment.

In one aspect, the invention comprises a railway digital twin module configured to maintain a continuously synchronized virtual representation of the physical railway infrastructure, the digital twin storing operational information associated with train movement, signal status, track occupancy, infrastructure health, communication status, and historical operational records for coordinated decision making.

In one aspect, the invention includes an intelligent multi-agent coordination module configured to distribute railway operational responsibilities among multiple intelligent railway agents, each agent executing one or more operational capabilities while exchanging operational information with other agents for coordinated railway control.

In one aspect, the system further comprises a capability health assessment module configured to independently evaluate each railway operational capability using communication quality, sensor reliability, processing latency, computational resource availability, and prediction confidence to generate a railway capability health index indicative of the operational condition of the corresponding capability.

In one aspect, the invention includes an adaptive capability reallocation module configured to identify degraded operational capabilities, evaluate candidate intelligent railway agents using operational health, capability compatibility, dependency satisfaction, and available computational resources, determine a capability reallocation score for each candidate agent, and dynamically migrate only the degraded capability to a selected compatible agent.

In one aspect, the system further comprises a railway safety validation module configured to evaluate the proposed capability reassignment against predefined railway operational constraints including route consistency, signal compatibility, interlocking conditions, track occupancy verification, collision safety, and associated operational dependencies, and to authorize capability migration only when the evaluated safety validation index satisfies a predefined safety criterion.

In one aspect, the invention further comprises a rollback and recovery module configured to automatically restore a previously validated capability configuration whenever the migrated capability fails to satisfy predefined operational or safety requirements after deployment, thereby preserving a verified safe operational state.

In one aspect, the present invention addresses the technical problem of maintaining continuous railway operation despite partial degradation of individual intelligent railway capabilities by introducing a capability-centric adaptive reallocation framework that selectively migrates only affected operational capabilities, validates the reconfigured architecture against railway-specific safety constraints, and restores previously validated configurations when necessary, without replacing entire intelligent railway agents.

In one aspect, the invention provides the technical advantage of improving operational resilience, fault tolerance, computational resource utilization, recovery efficiency, and safety assurance compared with conventional railway systems employing static redundancy or complete software-agent replacement mechanisms.

In one aspect, the invention enables scalable deployment across intelligent railway signalling systems, electronic interlocking systems, autonomous train control systems, railway traffic management platforms, and other distributed railway control infrastructures requiring adaptive, safety-critical, and uninterrupted operational intelligence

BRIEF DESCRIPTION OF THE DRAWINGS

The manner in which the present invention is formulated is given a more particular description below, briefly summarized above, may be had by reference to the components, some of which is illustrated in the appended drawing It is to be noted; however, that the appended drawing illustrates only typical embodiments of this invention and are therefore should not be considered limiting of its scope, for the system may admit to other equally effective embodiments.

Throughout the drawings, the same drawing reference numerals will be understood to refer to the same elements and features.

The features and advantages of the present invention will become more apparent from the following detailed description a long with the accompanying figures, which forms a part of this application and in which:

Figure 1 illustrates a process flow diagram of the proposed adaptive capability reallocation system for intelligent railway operational safety, depicting railway data acquisition, digital twin generation, intelligent multi-agent coordination, capability health assessment, adaptive capability reallocation, railway safety validation, rollback and recovery, and railway decision and monitoring.

Figure 2 illustrates an overall operational workflow of the proposed adaptive capability reallocation system, showing capability health evaluation, degradation detection, capability reallocation, railway safety validation, rollback and recovery, and continuation of railway operation based on the evaluated operational state.

REFERENCE NUMERALS

100 - adaptive capability reallocation system for intelligent railway operational safety

DETAILED DESCRIPTION OF THE INVENTION

The present invention will now be described in detail with reference to the accompanying drawing. The following description is provided to illustrate the invention and should not be construed as limiting the scope of the invention.

The present invention provides an adaptive capability reallocation system (100) for intelligent railway operational safety. The system (100) enables continuous, resilient, and fault-tolerant railway operation by dynamically monitoring the operational condition of individual railway capabilities executed by multiple intelligent railway agents and selectively reallocating only degraded capabilities without interrupting unaffected railway functions. Unlike conventional railway safety systems that replace an entire software agent or invoke static failover mechanisms after fault detection, the proposed system (100) independently evaluates the operational health of each railway capability, predicts partial degradation, identifies compatible intelligent railway agents, validates the proposed capability migration against predefined railway safety constraints,

and activates the validated configuration while maintaining uninterrupted railway operations. The system (100) further maintains capability dependency information, migration history, version control, and rollback mechanisms to ensure reliable and safe adaptive reconfiguration under dynamic railway operating conditions.

Figure 1 illustrates a block diagram of the adaptive capability reallocation system (100) for intelligent railway operational safety. The system (100) comprises a railway data acquisition module, a railway digital twin module, an intelligent multi-agent coordination module, a capability health assessment module, an adaptive capability reallocation module, a railway safety validation module, a rollback and recovery module, and a railway decision and monitoring module, which collectively enable adaptive capability management while maintaining continuous railway operation.

The railway data acquisition module continuously acquires operational information from railway infrastructure including railway signals, point machines, track circuits, axle counters, train positioning systems, train speed sensors, communication network status, voltage sensors, and environmental sensors. The acquired information is synchronized, validated, and converted into a real-time railway operational dataset representing the current operating condition of the railway network. The module further detects abnormal operating conditions and forwards the validated operational dataset to the railway digital twin module for further processing.

The railway digital twin module receives the validated operational dataset and continuously maintains a synchronized virtual representation of the physical railway infrastructure. The digital twin stores operational information including train positions, train movement, signal status, track occupancy, point machine position, communication network condition, railway infrastructure health, and historical operational records. By continuously updating the virtual railway environment, the module enables intelligent railway agents to evaluate present operating conditions and analyze possible operational scenarios before railway control decisions are executed. The synchronized digital twin further serves as a common operational environment shared among all intelligent railway agents.

The intelligent multi-agent coordination module distributes railway operational responsibilities among multiple intelligent railway agents. Each intelligent agent performs one or more specialized railway operational functions including signal monitoring, route planning, train movement supervision, interlocking verification, and railway safety validation. During execution, the intelligent agents continuously exchange operational information with one another to ensure coordinated railway decision-making and transmit execution status information to the capability health assessment module.

The capability health assessment module continuously evaluates the operational condition of every railway operational capability independently rather than assessing the health of an entire intelligent railway agent. The module monitors communication quality, sensor reliability, processing latency, computational resource availability, and prediction confidence associated with each operational capability. These runtime parameters are combined to compute a Railway Capability Health Index (RCHI), which represents the operational reliability of the corresponding capability. When the calculated Railway Capability Health Index falls below a predefined threshold, the corresponding capability is identified as degraded and is forwarded to the adaptive capability reallocation module for further processing.

The capability health assessment module combines the monitored operational parameters to compute a **Railway Capability Health Index (RCHI)** representing the operational reliability of the corresponding railway capability according to Equation (1):

where:

$$RCHI = w_1C + w_2S + w_3L + w_4R + w_5P$$

- **C** represents communication quality;
- **S** represents sensor reliability;
- **L** represents processing latency score;
- **R** represents resource availability;
- **P** represents prediction confidence;

- w_1–w_5 represent predefined weighting coefficients satisfying

The calculated Railway Capability Health Index ranges between zero and one, wherein a value closer to one indicates a healthy railway operational capability, while a value approaching zero indicates increasing degradation. When the Railway Capability Health Index falls below a predefined threshold, the corresponding capability is identified as degraded and is forwarded to the adaptive capability reallocation module.

The adaptive capability reallocation module represents the principal functionality of the system (100). Upon receiving information regarding a degraded capability, the module identifies the affected capability, searches for compatible intelligent railway agents, verifies dependency relationships, evaluates available computational resources, and computes a Capability Reallocation Score (CRS) for each candidate intelligent railway agent using parameters including operational health, capability compatibility, dependency satisfaction, and resource availability. The intelligent railway agent having the highest valid Capability Reallocation Score is selected as the destination agent for execution of the degraded capability. Unlike conventional systems that replace complete intelligent agents, the proposed system selectively reallocates only the affected railway capability while allowing all remaining healthy capabilities to continue normal operation.

The adaptive capability reallocation module computes a **Capability Reallocation Score (CRS)** for each candidate intelligent railway agent according to Equation (2):

$$CRS = \alpha H + \beta C + \gamma D + \delta A$$

where:

- **H** denotes health score;
- **C** denotes compatibility score;
- **D** denotes dependency satisfaction score;
- **A** denotes available resource score;

wherein the weighting coefficients satisfy

The intelligent railway agent having the highest valid Capability Reallocation Score is selected for execution of the degraded railway operational capability, subject to successful railway safety validation.

Before the migrated capability is activated, the railway safety validation module evaluates whether the proposed capability assignment satisfies predefined railway safety constraints. The validation process verifies route consistency, signal compatibility, interlocking constraints, track occupancy verification, and collision safety. Based on the number of satisfied safety constraints, the module computes a Safety Validation Index (SVI). Only when the evaluated Safety Validation Index satisfies the predefined safety requirement is the capability migration approved for execution. This validation prevents unsafe railway operation that may otherwise result from inappropriate capability reassignment.

If the proposed capability migration fails to satisfy the required safety conditions or subsequently fails after deployment, the rollback and recovery module automatically restores the previously validated capability configuration. The rollback process restores the previous capability assignment, reinstates operational dependencies, restores capability version information, and records operational logs associated with the migration event. Consequently, the railway system resumes operation using the last verified safe configuration without interrupting railway operation.

The railway decision and monitoring module receives validated railway operational decisions after successful capability migration or rollback recovery. The module continuously displays capability health, railway operational status, capability reallocation events, safety alerts, and operational history while transmitting validated railway control commands to railway signalling and control infrastructure. Thus, the block diagram of Figure 1 illustrates the coordinated operation of the various modules of the adaptive capability reallocation system (100), enabling capability-level fault tolerance, adaptive reconfiguration, and uninterrupted railway operation.

After evaluating the predefined railway safety constraints, the railway safety validation module computes a **Safety Validation Index (SVI)** according to Equation (3):

$$SVI = \frac{N_S}{N_t}$$

where:

- $\mathbf{N_s}$ represents the number of satisfied railway safety constraints; and
- $\mathbf{N_t}$ represents the total number of evaluated railway safety constraints.

A higher Safety Validation Index indicates greater compliance with the predefined railway safety requirements. Capability migration is approved only when the computed Safety Validation Index satisfies a predefined safety criterion; otherwise, the rollback and recovery module restores the previously validated railway operational configuration.

Figure 2 illustrates the overall operational workflow implemented by the adaptive capability reallocation system (100). The workflow begins with continuous acquisition of railway operational data from the physical railway infrastructure. The acquired operational information is supplied to the railway digital twin module, where a continuously synchronized virtual representation of the railway environment is generated and maintained for subsequent operational analysis.

The synchronized railway information is subsequently processed by the capability health assessment module, which evaluates the operational condition of each railway operational capability by computing the Railway Capability Health Index (RCHI). The computation utilizes communication quality, sensor reliability, processing latency, resource availability, and prediction confidence associated with each capability. The calculated Railway Capability Health Index is compared with a predefined capability health threshold to determine whether the corresponding railway capability is operating normally or has become degraded.

When the evaluated Railway Capability Health Index indicates that the capability remains healthy, the system (100) allows the corresponding railway operational capability to continue executing without interruption. Since no degradation is detected, capability migration is not initiated, and the railway operation proceeds under normal operating conditions while the system continues monitoring the health of all operational capabilities in real time.

When the Railway Capability Health Index indicates degradation of an operational capability, the adaptive capability reallocation module is activated. The module evaluates available intelligent railway agents by computing the Capability Reallocation Score (CRS) for each candidate agent using operational health, compatibility, dependency satisfaction, and computational resource availability. Based on the evaluated scores, the most suitable intelligent railway agent is selected for execution of the degraded operational capability.

Following candidate selection, the proposed capability assignment is forwarded to the railway safety validation module. The module evaluates railway-specific operational constraints including route consistency, signal compatibility, interlocking verification, track occupancy, and collision safety, and computes the Safety Validation Index (SVI). If the evaluated Safety Validation Index satisfies the predefined safety requirement, the migrated capability is approved, activated, and railway operation continues without interruption using the updated capability assignment.

Conversely, if the Safety Validation Index does not satisfy the required safety criterion, the proposed capability migration is rejected and the rollback and recovery module is activated. The rollback mechanism restores the previously validated capability assignment and re-establishes the previously verified railway operational configuration. Thereafter, railway operation resumes using the restored safe configuration while the system (100) continues continuous health monitoring for subsequent adaptive capability management. Accordingly, Figure 2 illustrates the complete operational workflow through which the proposed adaptive capability reallocation system (100) achieves capability-level degradation detection, intelligent capability reassignment, safety validation, rollback recovery, and uninterrupted railway operation under dynamic operating conditions.

n an alternative embodiment, the adaptive capability reallocation system (100) may be implemented with varying numbers of intelligent railway agents depending on the operational requirements of the railway network. Each intelligent railway agent may execute one or more railway operational capabilities including signal monitoring, route planning, train movement supervision, interlocking verification, and safety validation, wherein the capability health assessment module continuously evaluates the operational condition of each capability independently. The adaptive capability reallocation module selectively migrates only degraded

capabilities to compatible intelligent railway agents while unaffected capabilities continue normal execution, thereby maintaining uninterrupted railway operation.

In another embodiment, the railway data acquisition module may receive operational information from different combinations of railway infrastructure including signalling systems, point machines, track circuits, axle counters, train positioning systems, train speed sensors, communication networks, voltage monitoring units, and environmental sensors. The railway digital twin module continuously synchronizes the acquired operational information to provide a virtual representation of the railway infrastructure for capability health assessment, adaptive capability reallocation, railway safety validation, and rollback recovery without altering the overall operational principle of the system (100).

In the best mode of operation, the adaptive capability reallocation system (100) continuously acquires real-time operational information from railway infrastructure and synchronizes the acquired information within the railway digital twin to maintain an updated virtual representation of the physical railway environment. Multiple intelligent railway agents execute respective railway operational capabilities while the capability health assessment module continuously computes the Railway Capability Health Index (RCHI) using communication quality, sensor reliability, processing latency, resource availability, and prediction confidence associated with each capability.

When the Railway Capability Health Index of an operational capability falls below a predefined threshold, the adaptive capability reallocation module identifies compatible intelligent railway agents and computes the Capability Reallocation Score (CRS) to determine the most suitable destination agent. The proposed capability assignment is subsequently evaluated by the railway safety validation module against predefined railway safety constraints including route consistency, signal compatibility, interlocking conditions, track occupancy verification, and collision safety. Upon successful validation, only the degraded operational capability is migrated to the selected intelligent railway agent while all remaining healthy capabilities continue execution without interruption. If the migrated capability fails to satisfy the predefined operational or safety requirements, the rollback and recovery module automatically restores the previously validated capability configuration, thereby maintaining continuous, reliable, and safe railway operation.

The present invention provides a capability-centric adaptive railway operational architecture that selectively reallocates only degraded railway operational capabilities instead of replacing complete intelligent railway agents. By independently assessing capability health, dynamically selecting compatible intelligent railway agents, and validating each capability migration before activation, the invention minimizes unnecessary failover operations, reduces recovery time, improves computational resource utilization, and maintains uninterrupted railway operation during partial system degradation. Furthermore, the integration of railway data acquisition, digital twin synchronization, intelligent multi-agent coordination, capability health assessment, adaptive capability reallocation, and railway safety validation provides a unified framework for resilient railway operation.

The invention further enhances operational safety by incorporating railway-specific validation and automatic rollback mechanisms that prevent unsafe capability migration and restore previously verified operational configurations whenever predefined safety requirements are not satisfied. Consequently, the invention improves operational reliability, fault tolerance, system availability, scalability, and overall resilience of intelligent railway signalling, electronic interlocking, train supervision, and railway traffic management systems operating under dynamic and partially degraded conditions.

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Experimental Validation Results:

The proposed adaptive capability reallocation system (100) was evaluated under multiple railway operational scenarios to validate capability health assessment, adaptive capability reallocation, railway safety validation, and rollback recovery. The evaluation utilized operational information acquired from railway infrastructure including signalling systems, track circuits, point machines, communication networks, and intelligent railway agents. The system computed a Railway Capability Health Index (RCHI), Capability Reallocation Score (CRS), and Safety Validation Index (SVI) to determine the operational state of individual railway capabilities and the corresponding adaptive response.

Under a normal railway operating condition, the acquired operational information indicated that the signalling system, track occupancy, point machines, communication network, and intelligent railway agents were functioning within acceptable operating limits. The capability health assessment module computed a Railway Capability Health Index (RCHI) of **0.811**, which exceeded the predefined capability health threshold of **0.75**. Accordingly, the evaluated railway capability was classified as healthy, adaptive capability reallocation was not initiated, and uninterrupted railway operation continued using the existing intelligent railway agent. The obtained results demonstrate that the proposed system (100) maintains stable railway operation while avoiding unnecessary capability migration during minor operational fluctuations.

A subsequent evaluation was performed by introducing degradation in the communication capability while maintaining acceptable values for sensor reliability, processing latency, computational resource availability, and prediction confidence. Under these conditions, the capability health assessment module computed a Railway Capability Health Index (RCHI) of **0.723**, which was below the predefined threshold. Consequently, the adaptive capability reallocation module evaluated available intelligent railway agents using the Capability Reallocation Score (CRS) and selected the candidate agent having the highest score of **0.940**. Following verification of route consistency, signal compatibility, interlocking conditions, track occupancy, collision safety, and associated operational constraints, the Safety Validation Index (SVI) was computed as **1.00**, satisfying the predefined safety requirement. The degraded railway operational capability was therefore successfully migrated to the selected intelligent railway agent without interrupting railway operation or requiring rollback recovery.

A further evaluation was conducted under severe operational degradation by simultaneously reducing communication quality, sensor reliability, computational resource availability, prediction confidence, and processing performance while introducing unsafe railway operating conditions. The capability health assessment module computed a Railway Capability Health Index (RCHI) of **0.170**, indicating severe capability degradation. Although the adaptive capability reallocation module identified a candidate intelligent railway agent with the highest Capability Reallocation Score (CRS) of **0.940**, the railway safety validation module determined that multiple predefined safety constraints were not satisfied, resulting in a Safety Validation Index (SVI) of **0.00**, which was below the required safety threshold. Consequently, the proposed capability migration was

rejected, and the rollback and recovery module automatically restored the previously validated capability assignment and safe railway operational configuration.

The experimental evaluation demonstrates that the proposed adaptive capability reallocation system (100) accurately distinguishes healthy and degraded railway operational capabilities, selectively reallocates only degraded capabilities when operational and safety requirements are satisfied, and automatically restores previously validated configurations whenever safety validation fails. The obtained results validate the capability-centric adaptive reallocation architecture for improving railway operational resilience, fault tolerance, computational resource utilization, recovery efficiency, and uninterrupted railway operation under dynamic operating conditions.

CLAIMS

I/We Claim:

1. A computer-implemented adaptive capability reallocation system (100) for intelligent railway operational safety, the system (100) comprising:

a railway data acquisition module configured to acquire operational information from railway infrastructure including signalling systems, point machines, track circuits, train positioning systems, communication networks and environmental sensors;

a railway digital twin module configured to generate and continuously update a synchronized virtual representation of the railway infrastructure based on the acquired operational information;

an intelligent multi-agent coordination module configured to distribute railway operational capabilities among a plurality of intelligent railway agents;

a capability health assessment module configured to independently evaluate an operational condition of each railway operational capability by computing a Railway Capability Health Index (RCHI) using communication quality, sensor reliability, processing latency, computational resource availability and prediction confidence;

an adaptive capability reallocation module configured to identify a degraded railway operational capability when the Railway Capability Health Index satisfies a predefined degradation criterion, determine a plurality of compatible intelligent railway agents, compute a Capability Reallocation Score (CRS) for each compatible intelligent railway agent using operational health, capability compatibility, dependency satisfaction and available computational resources, and select an intelligent railway agent based on the computed Capability Reallocation Score;

a railway safety validation module configured to evaluate a proposed capability assignment using predefined railway safety constraints including route consistency, signal compatibility, interlocking constraints, track occupancy verification and collision safety, and compute a Safety Validation Index (SVI); and

a rollback and recovery module configured to restore a previously validated capability assignment when the computed Safety Validation Index does not satisfy a predefined safety criterion or when a migrated capability fails to satisfy predefined operational requirements after deployment.

2. The system (100) as claimed in claim 1, wherein the railway data acquisition module synchronizes operational information received from railway signals, point machines, track circuits, axle counters, train positioning systems, train speed sensors, communication network status, voltage sensors and environmental sensors to generate a validated railway operational dataset for the railway digital twin module.
3. The system (100) as claimed in claim 1, wherein the railway digital twin module continuously maintains operational information comprising train position, train movement, signal status, track occupancy, point machine position, communication network status, railway infrastructure health and historical operational records for coordinated capability execution.
4. The system (100) as claimed in claim 1, wherein the capability health assessment module computes the Railway Capability Health Index using weighted communication quality, sensor reliability, processing latency, computational resource availability and prediction confidence associated with each railway operational capability, and independently evaluates each railway operational capability without assessing an entire intelligent railway agent.
5. The system (100) as claimed in claim 1, wherein the adaptive capability reallocation module performs capability identification, compatible intelligent railway agent selection, dependency verification, resource evaluation and capability reassignment by ranking candidate intelligent railway agents according to the computed Capability Reallocation Score.
6. The system (100) as claimed in claim 1, wherein the adaptive capability reallocation module selectively reallocates only the degraded railway operational capability while maintaining execution of remaining healthy railway operational capabilities by the corresponding intelligent railway agents.
7. The system (100) as claimed in claim 1, wherein the railway safety validation module independently evaluates route consistency, signal compatibility, interlocking

constraints, track occupancy verification and collision safety to compute the Safety Validation Index before activation of the selected railway operational capability.

8. The system (100) as claimed in claim 1, wherein the rollback and recovery module restores a previously validated capability configuration together with dependency relationships, capability version information and migration history upon unsuccessful capability validation or capability failure after migration.

9. The system (100) as claimed in claim 1, further comprising a railway decision and monitoring module configured to receive validated railway operational decisions, generate railway control commands, monitor capability health, capability migration events, railway operational status and safety alerts, and maintain operational history associated with capability execution.

10. A computer-implemented method for adaptive capability reallocation in intelligent railway operational safety using the system (100) as claimed in any one of the preceding claims, the method comprising:

acquiring railway operational information from railway infrastructure;

generating and updating a synchronized railway digital twin;

allocating railway operational capabilities among intelligent railway agents;

independently computing a Railway Capability Health Index for each railway operational capability;

identifying degraded railway operational capabilities based on the computed Railway Capability Health Index;

computing Capability Reallocation Scores for compatible intelligent railway agents;

selecting an intelligent railway agent based on the computed Capability Reallocation Scores;

validating the selected capability assignment by computing a Safety Validation Index using predefined railway safety constraints;

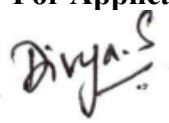
activating the selected capability assignment when the Safety Validation Index satisfies a predefined safety criterion; and

restoring a previously validated capability configuration when the Safety Validation Index fails to satisfy the predefined safety criterion or when post-deployment operational requirements are not satisfied.

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Date: 23.08.2026

Name and Signature of the Patent Agent
For Applicant



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For Daniel & Gladys.

A Reconfigurable FPGA-Based Reflex Training System for Hockey Goalkeepers

ABSTRACT

An adaptive capability reallocation system (100) for intelligent railway operational safety comprises a railway data acquisition module, a railway digital twin module, an intelligent multi-agent coordination module, a capability health assessment module, an adaptive capability reallocation module, a railway safety validation module, a rollback and recovery module, and a railway decision and monitoring module. The railway data acquisition module acquires operational information from railway infrastructure, while the railway digital twin module maintains a synchronized virtual representation of the railway environment. The capability health assessment module computes a Railway Capability Health Index for individual railway operational capabilities. The adaptive capability reallocation module determines compatible intelligent railway agents using a Capability Reallocation Score and selectively reallocates degraded operational capabilities. The railway safety validation module computes a Safety Validation Index before capability activation, and the rollback and recovery module restores a previously validated capability configuration when predefined operational requirements are not satisfied.

Fig 1

Date: 23.08.2026

Name and Signature of the Patent Agent
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Ms.Divya Sridharan
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